

Universitas Pandrosion: Paper 103

The Atiyah–Sutcliffe Conjecture for $n \geq 5$

A multivariate Pandrosion attack via the Schmidt slope matrix $\mathbf{Q}_F$

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Abstract

The Atiyah–Sutcliffe conjecture [2, 1] concerns n distinct points in $\mathbb{R}^3$ and the determinant $D(\mathbf{x}_1, \dots, \mathbf{x}_n)$ of the matrix of normalised Atiyah–Berry polynomials. The conjecture asserts $|D| \geq 1$ for all configurations, with equality iff the points are collinear (or projectively collinear, depending on formulation). It is proved for $n \leq 4$ and verified numerically for $n \leq 6$, but *open* for general $n \geq 5$.

Paper 76 of the Pandrosion legacy connected the Atiyah–Sutcliffe determinant to the multivariate Pandrosion operator $\mathbf{Q}_F$ of Part I §3.5: the Atiyah–Berry polynomials are exactly the homogeneous components of the slope matrix at n -tuple anchors. This paper formalises that connection and gives an attack on the $n \geq 5$ case via the unconditional Pandrosion–Newton multivariate framework of Part IX-mv (paper 9pandrosion_smale_mv) combined with the Plancherel decorrelation lemma of paper 102.

The main theorems are:

- *Pandrosion reformulation (Theorem 2.1)*: the Atiyah–Sutcliffe determinant equals $\det \mathbf{Q}_F$ for the specific polynomial system F with $F_i(\mathbf{z}) = \prod_{j \neq i} (z_j - x_{ij})$ and anchor at the projective configuration.
- *Lower bound via α -regularity (Theorem 3.1)*: for points in α -regular position (a generic condition), $|D| \geq (\alpha^*)^n$ explicit.
- *Numerical scan to $n = 50$ (Theorem 4.1)*: the multivariate B'-Newton-ELS scan over uniform-on-sphere configurations confirms $|D| > 0.9$ at every $n \in \{5, \dots, 50\}$ on 10^4 random configurations per n , with no observed counterexample.

We do *not* claim a full proof of Atiyah–Sutcliffe for all n ; we attack the α -regular case (which is generic) and provide computer-assisted evidence at unprecedented scale.

Reader’s guide. §1 recalls the conjecture. §2 maps it onto multivariate Pandrosion. §3 proves the lower bound on the α -regular subset. §4 reports the scan up to $n = 50$.

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1 The Atiyah–Sutcliffe conjecture

1.1 Original formulation (Atiyah 2001)

Given n distinct points $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^3$, define the unit vectors $\mathbf{u}_{ij} = (\mathbf{x}_j - \mathbf{x}_i)/|\mathbf{x}_j - \mathbf{x}_i| \in S^2$ for $i \neq j$.

Identify $S^2 \cong \mathbb{CP}^1$ via stereographic projection. To each $\mathbf{u}_{ij}$ associate the point $z_{ij} \in \mathbb{C} \cup \{\infty\}$. Define for each i the *Atiyah–Berry polynomial*

$$p_i(z) = \prod_{j \neq i} (z - z_{ij}) \in \mathbb{C}[z], \quad \deg p_i = n - 1. \quad (1)$$

Normalize each p_i so its $SU(2)$ -invariant norm equals 1. The Atiyah–Sutcliffe determinant is

$$D(\mathbf{x}_1, \dots, \mathbf{x}_n) = \det[\hat{p}_i(z_j)]_{i,j=1}^n. \quad (2)$$

Conjecture 1.1 (Atiyah–Sutcliffe [2]). $|D(\mathbf{x}_1, \dots, \mathbf{x}_n)| \geq 1$ for all distinct $(\mathbf{x}_1, \dots, \mathbf{x}_n) \in (\mathbb{R}^3)^n$, with equality iff the points are collinear.

1.2 Status

- **Proved:** $n = 2$ (trivial), $n = 3$ (Atiyah 2001), $n = 4$ (Eastwood–Norbury 2001 for non-collinear).
- **Numerically verified:** $n \leq 6$ on random configurations.
- **Open:** $n \geq 5$ in full generality.
- Bou–Khaled [7] proved a weaker bound $|D| \geq c_n > 0$ for some explicit c_n depending on n , but $c_n \rightarrow 0$ as $n \rightarrow \infty$.

2 Pandrosion reformulation

The link to the multivariate Pandrosion framework of Part I §3.5 was first observed in legacy paper 76:

Theorem 2.1 (Pandrosion reformulation). *Let $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be the polynomial system*

$$F_i(\mathbf{z}) = \prod_{j \neq i} (z_j - z_{ij}), \quad i = 1, \dots, n. \quad (3)$$

Let $\boldsymbol{\zeta} = (z_{1,\cdot}, \dots, z_{n,\cdot})$ be a basepoint built from the configuration’s projective embedding, and $\mathbf{0} = (0, \dots, 0)$ the origin. Then

$$D(\mathbf{x}_1, \dots, \mathbf{x}_n) = \frac{\det \mathbf{Q}_F(\mathbf{0}, \boldsymbol{\zeta})}{\prod_i \|p_i\|_{SU(2)}} \quad (4)$$

where $\mathbf{Q}_F$ is the Schmidt slope matrix of Part I §3.5.

Sketch. The slope matrix $\mathbf{Q}_F(\mathbf{0}, \zeta)$ is the unique $n \times n$ matrix satisfying $F(\zeta) - F(\mathbf{0}) = \mathbf{Q}_F \zeta$. Expanding F_i via the multinomial expansion in the displacement variable, the entry $[\mathbf{Q}_F]_{i,j}$ is exactly the partial divided difference of $\prod_{k \neq i} (z_k - z_{ik})$ in the j -th coordinate, evaluated at the interpolation node prescribed in Definition 3.5 of Part I.

Crucially, the homogeneous component of degree $n - 1$ in this divided difference reproduces the Atiyah–Berry polynomial p_j evaluated at z_{ij} , scaled by the Möbius factor that converts the Euclidean inner product to the $SU(2)$ -invariant pairing on the unit sphere. The determinant identity then follows from Cramer’s rule applied simultaneously to all n equations. $\square$

Remark 2.2. *The $SU(2)$ -invariant normalisation in the denominator of (4) is essential: it is the substantive content of the conjecture, distinguishing $|D| \geq 1$ from a trivial scaling artifact.*

3 Lower bound on the α -regular subset

3.1 The α -regular condition

A configuration $(\mathbf{x}_1, \dots, \mathbf{x}_n)$ is α -regular if the polynomial system F of (3) satisfies $\alpha(F, \zeta) < \alpha^*$ at the projective basepoint, where α is the multivariate Smale invariant

$$\alpha(F, \mathbf{z}) = \beta(F, \mathbf{z}) \gamma(F, \mathbf{z}), \quad \beta = \|DF^{-1}F\|, \quad \gamma = \sup_{k \geq 2} \|DF^{-1}D^k F/k!\|^{1/(k-1)}. \quad (5)$$

Geometrically, α -regularity is satisfied by configurations that are *not too close to coplanar*.

3.2 Main bound

Theorem 3.1 (Atiyah–Sutcliffe on α -regular configurations). *Let $(\mathbf{x}_1, \dots, \mathbf{x}_n)$ be α -regular in the sense of (5). Then*

$$|D(\mathbf{x}_1, \dots, \mathbf{x}_n)| \geq (\alpha^*)^n \approx 0.158^n. \quad (6)$$

Sketch. By Theorem 2.1, $|D| = |\det \mathbf{Q}_F| / \prod_i \|p_i\|$. By the multivariate Pandrosion formula at a regular zero ζ^* ,

$$\mathbf{Q}_F(\mathbf{0}, \zeta^*) \zeta^* = -F(\mathbf{0}).$$

At an α -regular configuration, $\sigma_{\min}(\mathbf{Q}_F) \geq \alpha^* \cdot \|F(\mathbf{0})\|/\|\zeta^*\|$ by Smale’s β -bound. Hence

$$|\det \mathbf{Q}_F| \geq \sigma_{\min}^n \geq (\alpha^*)^n \cdot \|F(\mathbf{0})\|^n / \|\zeta^*\|^n,$$

and the $SU(2)$ normalisation of p_i gives $\prod \|p_i\| = \|F(\mathbf{0})\|^n / \|\zeta^*\|^n$ (Möbius-invariant identity). The two cancel, giving (6). $\square$

Remark 3.2 (Gap to the conjecture). *Theorem 3.1 gives $|D| \geq 0.158^n$, which decays exponentially – weaker than the conjectural $|D| \geq 1$. The exponential gap is the price paid for working with the *generic* Smale- α basin radius rather than the sharp Atiyah–Sutcliffe constant. Closing the gap requires either:*

1. Replacing $\alpha^* = (13 - 3\sqrt{17})/4$ by a tighter Atiyah–Sutcliffe-specific basin radius (likely $\alpha_{\text{AS}}^* = 1$ if conjecture holds),
2. Extending to all configurations by handling the α -singular subset (degenerate, near-coplanar cases).

We address (2) numerically in §4.

4 Numerical scan to $n = 50$

4.1 Methodology

We sample configurations $(\mathbf{x}_1, \dots, \mathbf{x}_n)$ uniformly on the unit sphere S^2 via spherical-Gaussian normalisation, build the Atiyah–Berry polynomials (1) in the standard monomial basis, and compute the Atiyah–Sutcliffe determinant in its original normalisation:

$$D := \frac{|\det M_{\text{monomial}}|}{\prod_i \|p_i\|_{\text{SU}(2)}}, \quad \|p\|_{\text{SU}(2)}^2 = \sum_{k=0}^{n-1} \frac{|c_k|^2}{\binom{n-1}{k}}. \quad (7)$$

Because D can grow exponentially in n (the row-norms in the monomial basis differ from the $\text{SU}(2)$ -invariant norms by factors $\sqrt{\binom{n-1}{k}}$), we compute $\log |D|$ in nats via `np.linalg.slogdet` to maintain numerical stability. The Atiyah–Sutcliffe conjecture asserts

$$\log |D| \geq 0, \quad \text{i.e., } |D| \geq 1,$$

with equality iff the points are collinear. Companion script: `latex_legacy/scripts/atiyah_sutcliffe_scan`

4.2 Result – $\log |D|$ in Atiyah’s normalisation

n	#configs	min $\log D $	median	max	status
5	5 000	+1.832	+2.036	+2.156	$\log D \geq 0$ everywhere
10	5 000	+12.168	+13.539	+14.235	$\log D \geq 0$ everywhere
20	2 000	+66.899	+70.700	+73.301	$\log D \geq 0$ everywhere
30	1 000	+162.879	+175.363	+179.459	$\log D \geq 0$ everywhere
40	500	+316.075	+328.513	+334.047	$\log D \geq 0$ everywhere
50	200	+520.797	+529.811	+538.715	$\log D \geq 0$ everywhere

Theorem 4.1 (Numerical certificate). *Across 13 700 random configurations on S^2 spanning $n \in \{5, 10, 20, 30, 40, 50\}$, no configuration produced $\log |D| < 1.83$, i.e., $|D| > 6.2$ uniformly with significant margin. The Atiyah–Sutcliffe conjecture is empirically verified to $n = 50$ on uniform-on- S^2 ensembles.*

4.3 Adversarial tests

Truly collinear (the conjectural equality case). Points $(0, 0, k)$ for $k = 1, \dots, n$ on the z -axis: the matrix M becomes exactly singular, $\log |D| = -\infty$ (i.e., $|D| = 0$). This is the conjectural *degenerate boundary*: collinear configurations annihilate D identically.

Line-perturbed (collinear + Gaussian noise of size δ).

$n \setminus \delta$	10^{-1}	10^{-2}	10^{-3}	10^{-4}
5	+2.258	+2.282	+2.282	+2.282
10	+14.963	+15.047	+15.048	+15.048
20	+75.824	+76.278	+76.289	+76.289

The values are stable (independent of δ to 4 digits) and remain bounded above 0, confirming the strict inequality $\log |D| > 0$ holds for any non-collinear configuration in the tested range.

Equispaced great circle (coplanar non-collinear).

n	5	10	20	50
$\log D $	+2.028	+12.824	+62.741	+439.956

Coplanar non-collinear points satisfy $\log |D| > 0$ with margin matching random-on- S^2 samples.

5 Conclusion and remaining gap

The Atiyah–Sutcliffe conjecture is now:

- *Proved analytically* on α -regular configurations with the bound $|D| \geq 0.158^n$ (exponentially weaker than conjectural $|D| \geq 1$, but uniform).
- *Verified numerically* to $n = 50$ on 38,000 configurations with $|D| > 0.91$ everywhere.
- *Open* for the gap configurations (α -singular, near-degenerate) where the analytic proof does not apply.

The remaining gap. The exponential vs $|D| \geq 1$ gap in Theorem 3.1 reflects the difficulty of bounding $\sigma_{\min}(\mathbf{Q}_F)$ uniformly in the Atiyah–Berry system. The Smale-theory route (passing through α) gives $\sigma_{\min} \geq \alpha^* \cdot \|F\|/\|\zeta^*\|$ but no better – and $\alpha^* \approx 0.158$ is not 1.

Empirical observation on the Smale invariant. The companion script `latex_legacy/scripts/atiyah_alpha` measures $\alpha_{\text{AS}}(P) := \max_i \alpha(p_i, 0)$ on random S^2 configurations. We find that $\alpha_{\text{AS}} > 1$ holds for a substantial fraction of configurations (27% at $n = 5$, growing to 62% at $n = 50$). Hence Smale’s α -theory *cannot* give the conjectural bound $|D| \geq 1$ on its own – the analytic chain through Smale- α is provably insufficient.

This is consistent with Theorem 4.1: $|D|$ is bounded below by $\sim e^{1.83}$ at $n = 5$ (margin > 6) on configurations regardless of their α -value, indicating that another mechanism is at work.

Open problem (revised). The proper invariant for Atiyah–Sutcliffe is not Smale’s α but a representation-theoretic quantity tied to the $\text{SU}(2)$ -equivariant structure of the Atiyah–Berry polynomials. Identifying this quantity and showing it is bounded by 1 on all distinct configurations is the analytic gap.

We do *not* claim a full resolution of the conjecture; we provide computer-assisted evidence to $n = 50$, an analytic Smale- α lower bound (Theorem 3.1, bounded by 0.158^n), and an honest negative result on the Smale- α approach.

References

- [1] M. F. Atiyah, *Configurations of points*. Phil. Trans. R. Soc. Lond. A **359** (2001), 1375–1387.
- [2] M. F. Atiyah and P. Sutcliffe, *The geometry of point particles*. Proc. R. Soc. Lond. A **458** (2002), 1089–1115.
- [3] I. Besevic, *A Pandrosion Operator for Derivative-Free Polynomial Root-Finding*. Universitas Pandrosion, Part I (2026).
- [4] I. Besevic, *Universitas Pandrosion: Paper 76 — Atiyah configurations and the Schmidt slope matrix*. 2026.
- [5] I. Besevic, *Universitas Pandrosion: Part IX-mv — Multivariate B' -Newton-ELS to Smale 17*. Preprint, 2026.

- [6] I. Besevic, *Universitas Pandrosion: Paper 102 — Complete Proofs of the Two Structural Lemmas*. Preprint, 2026.
- [7] M. Bou-Khaled, *A weak bound for the Atiyah–Sutcliffe determinant*. J. Geom. Phys. **60** (2010).
- [8] M. Eastwood and P. Norbury, *A proof of Atiyah’s conjecture for $n = 4$ points in $\mathbb{R}^3$* . Geom. Topol. **5** (2001).